

SYNTHESIS-GROUNDED GENERATIVE DESIGN OF IONIZABLE LIPIDS

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ABSTRACT

Many ionizable-lipid discovery platforms define candidates by combining predefined heads, linkers and hydrophobic components, which preserves experimental tractability but fixes the molecular vocabulary before design begins. Whole-molecule generators can leave that vocabulary, yet they do not by themselves identify how newly generated component chemistry can be assembled or prepared. Here we introduce FORGE, a synthesis-grounded generative framework that represents each design as a complete molecular graph coupled to a reaction program. FORGE uses precursor roles, reaction-core positions and program depth as semantic coordinates while generating every atom and bond without component identifiers or fragment tokens. Across three ionizable-lipid reaction programs, FORGE achieved exact level-one (L1) transform-consistent yields of 78.7%, 77.5% and 31.3% for Ugi, repeated aza-Michael addition and repeated reductive amination, respectively. Relative to null generation followed by post-hoc filtering, the corresponding improvements were 67.24, 73.49 and 18.10 percentage points. Against an exact train-only catalogue assembler, FORGE generated open-ended exact-L1 products whose recovered precursor identities were outside the supplied catalogue, whereas this quantity was structurally zero for the catalogue method. In a bounded, route-blinded Ugi shortlist, 19 of 255 admitted products closed to complete computational synthesis dossiers; the remaining products were explicitly unresolved or search-censored rather than called unsynthesizable.

1 INTRODUCTION

Lipid nanoparticles (LNPs) have enabled the systemic delivery of nucleic-acid therapeutics, including small interfering RNA, mRNA vaccines and *in vivo* genome-editing reagents (Adams et al., 2018; Polack et al., 2020; Baden et al., 2021; Gillmore et al., 2021). Within each LNP, the ionizable lipid supports nucleic-acid complexation and influences particle assembly, endosomal escape and tissue-dependent delivery, making its molecular structure a central determinant of LNP function (Cullis & Hope, 2017; Hou et al., 2021). Ionizable lipids combine a protonatable headgroup with hydrophobic domains that vary in chain length, branching, unsaturation, asymmetry, linker chemistry and degradability (Semple et al., 2010; Jayaraman et al., 2012; Whitehead et al., 2014; Hajj et al., 2019; Sabnis et al., 2018; Lee et al., 2021). Small changes to these structural components can substantially alter delivery, yet their effects depend on the complete molecular context and do not consistently follow simple design rules (Cheng et al., 2020; Han et al., 2021; Hatit et al., 2023). Ionizable-lipid discovery therefore commonly relies on preparing modular chemical libraries and screening the resulting LNPs experimentally (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024).

Machine-learning models can accelerate this workflow by predicting LNP performance and prioritizing candidates before synthesis (Li et al., 2024; Xu et al., 2024; Wang et al., 2024). Existing generative approaches often incorporate synthesis either by restricting generation to predefined components or by evaluating synthesizability after generation (Gao & Coley, 2020; Gao et al., 2022; Koziarski et al., 2024). Restricting generation in this way can preserve an explicit assembly path, but it limits the search space to the structures that can be constructed from the supplied precursor inventory. With post-hoc filtering, the model generates a complete lipid without knowing which regions must behave as particular precursors or which bonds the intended reaction must form. The filter can only accept or reject the finished molecule.

With reaction conditioning, the model can coordinate the structure assigned to each precursor role, the reactive handles required by that role, the atoms surrounding the reaction core, the bond-forming event that joins the precursors, and the structures of different regions within the same complete lipid. The central question we address is: how can reaction knowledge guide de novo ionizable-lipid generation without restricting the model to a predefined component vocabulary? FORGE instead uses reaction programs to guide molecular generation without making predefined components the model’s vocabulary. In FORGE, a reaction program specifies which precursor roles participate, where bonds are formed or changed, and how transformations are ordered, but not the molecular identities of the precursors. Each generated product is then tested by decomposing it into role-compatible precursors and replaying the specified reaction to determine whether the original molecular graph is recovered exactly. If a required precursor is not directly available, FORGE searches for a supported reaction that can prepare it from other precursors and repeats this process until the route reaches available starting materials or no supported route remains. Designs with an unsupported preparation step are excluded from the route-complete set. To test whether this approach generalizes across reaction programs, we evaluate FORGE using three ionizable-lipid assembly chemistries: an AGILE-type amine–aldehyde–isocyanide Ugi three-component reaction, repeated aza-Michael addition, and repeated reductive amination. We make three main contributions:

1. **A reaction-program formulation of synthesis-grounded molecular generation.** We formalize assembly chemistry through precursor roles, reaction-core positions and transformation order while leaving precursor identities unspecified. This makes reaction compatibility an explicit modeling target without reducing generation to selection from a finite component inventory. Reaction programs encode assembly semantics, not experimental synthesis success.
2. **A reaction-guided whole-lipid and synthesis-program framework.** FORGE generates complete lipid graphs at atom and bond resolution, verifies the final assembly through exact decomposition and forward replay, and recursively constructs preparation routes for generated precursors. The resulting computational dossier connects accepted designs to supported starting materials and explicitly rejects incomplete routes.
3. **Empirical validation across reaction programs and component inventories.** Across three ionizable-lipid assembly chemistries, conditioning improved exact-L1 yield over matched null generation plus post-hoc filtering by 67.24, 73.49 and 18.10 percentage points. FORGE produced verifier-relative component novelty and nonzero open-ended exact-L1 yield that a train-only finite-component assembler cannot attain by construction. Bounded recursive route assessment closed 19 of 255 admitted Ugi shortlist products and explicitly retained unresolved and search-censored outcomes. These experiments test generalization across specified reaction programs, not zero-shot support for arbitrary chemistry.

2 PRELIMINARIES

2.1 IONIZABLE-LIPID STRUCTURE AND MODULAR ASSEMBLY

The structure of an ionizable lipid is typically organized into three functional regions: a protonatable headgroup, a linker, and one or more hydrophobic domains (Semple et al., 2010; Jayaraman et al., 2012). Together, these regions determine the ionization, stability, lipid packing, and membrane interactions of the molecule. Ionizable-lipid libraries are often synthesized by applying a common reaction to different combinations of precursor molecules (Akinc et al., 2008; Love et al., 2010; Li et al., 2024; Xu et al., 2024). Each precursor contributes a defined portion of the final lipid and contains the reactive group required for the assembly. The reaction forms the bonds that connect these precursor-derived regions. As a result, the assembly reaction determines how the regions are connected, while the precursor structures determine the molecular identity of the resulting lipid.

2.2 ASSEMBLY CHEMISTRIES STUDIED

We study three ionizable-lipid assembly chemistries: an AGILE-type amine–aldehyde–isocyanide Ugi three-component reaction, repeated aza-Michael addition, and repeated reductive amination. In the AGILE-type Ugi reaction, an amine head reacts with an aldehyde component and an isocyanide

component to form an α -amino amide that connects the headgroup to two hydrophobic regions (Xu et al., 2024). A second assembly strategy uses repeated aza-Michael additions to attach acrylate-bearing hydrophobic components sequentially to reactive amines in the headgroup (Li et al., 2023). Reductive amination sequentially couples aldehyde-bearing hydrophobic components to reactive amines in the headgroup through new carbon–nitrogen bonds (Xue et al., 2024).

2.3 REACTION TRANSFORMS AND EXACT ASSEMBLY

Each assembly chemistry defines a forward transformation that maps role-assigned precursor molecules to the product graphs permitted by that reaction. The corresponding reverse transformation decomposes a complete lipid into the precursor assignments that could have produced it. To verify a decomposition, we computationally replay the reaction on the recovered precursors. The product passes the level-one (L1) assembly check only if the reconstruction matches the original lipid exactly.

2.4 DISCRETE MOLECULAR GRAPH SPACES

Let $\mathcal{A}$ denote a finite vocabulary of atom states and $\mathcal{B}$ a finite vocabulary of bond states. A molecular graph $G = (V, E)$ assigns an atom state $a_i \in \mathcal{A}$ to each node $i \in V$ and a bond state $b_{ij} \in \mathcal{B}$ to each edge $(i, j) \in E$. For a fixed maximum heavy-atom count $N_{\max}$, let $\mathcal{G}$ denote the resulting finite space of labeled molecular graphs.

2.5 DISCRETE FLOW MATCHING

Let $\mathcal{X}$ be a finite state space, p_0 a tractable source distribution, and p_1 the target data distribution. Discrete flow matching constructs a time-indexed probability path $(p_t)_{t \in [0,1]}$ between these distributions, such that $p_{t=0} = p_0$ and $p_{t=1} = p_1$ (Qin et al., 2025). Given a target sample $x_1 \sim p_1$, the linear conditional path is

$$p_t(x_t | x_1) = (1 - t)p_0(x_t) + t\delta_{x_1}(x_t), \quad t \in [0, 1], \quad (1)$$

where δ_{x_1} denotes a point mass at x_1 . Marginalizing over x_1 gives

$$p_t(x) = (1 - t)p_0(x) + tp_1(x), \quad (2)$$

so the states remain discrete while their probabilities evolve continuously in time. A continuous-time Markov chain realizes this probability path through transition rates $R_t(x, y) \geq 0$, where $R_t(x, y)$ is the instantaneous rate of moving from state x to state y . The resulting distribution evolves according to

$$\frac{d}{dt}p_t(x) = \sum_{y \neq x} [p_t(y)R_t(y, x) - p_t(x)R_t(x, y)], \quad (3)$$

which balances the probability flowing into and out of each state. Given an intermediate state x_t , a neural network $p_\theta(x_1 | x_t, t)$ predicts the target state x_1 . We train this predictor by minimizing the denoising cross-entropy

$$\mathcal{L}_{\text{DFM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1), x_1 \sim p_1, x_t \sim p_t(\cdot | x_1)} [-\log p_\theta(x_1 | x_t, t)]. \quad (4)$$

For each endpoint x_1 , let $R_t(x, y | x_1)$ realize the conditional path above. We obtain the learned transition rate by averaging these conditional rates under the model prediction:

$$R_t^\theta(x, y) = \sum_{x_1 \in \mathcal{X}} p_\theta(x_1 | x, t) R_t(x, y | x_1). \quad (5)$$

Sampling draws $x_0 \sim p_0$ and evolves the state from $t = 0$ to $t = 1$ according to the learned transition rates R_t^θ .

3 FORGE

3.1 OVERVIEW

We formulate ionizable-lipid design as conditional generation under a specified reaction program, while leaving precursor identities unspecified. Let $G \in \mathcal{G}$ denote a complete ionizable-lipid graph

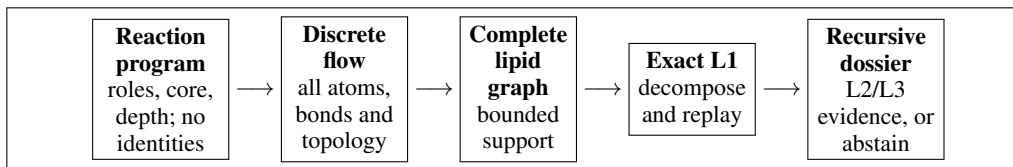


Figure 1: **FORGE generates complete ionizable-lipid graphs within reaction-program coordinates.** The reaction program specifies how molecular regions participate in an assembly without choosing their structures from a finite inventory. FORGE generates the complete molecular graph, verifies that the specified reaction reconstructs it, and recursively expands unavailable precursors until every branch terminates in evidence-qualified starting materials or the framework abstains. Exact replay and route closure are computational checks and do not establish experimental synthesis success.

and $P \in \mathcal{P}$ a reaction program specifying the assembly chemistry, precursor roles, reaction-core positions, and transformation order. FORGE models the conditional distribution $p_\theta(G | P)$ without access to precursor identifiers, stored component graphs, or fragment tokens. FORGE is organized into four stages:

1. **Reaction-program specification.** P defines the assembly chemistry, precursor roles, reaction-core positions, transformation order, and coarse per-role morphology, but not precursor identities.
2. **Whole-graph generation.** A conditional discrete flow generates every atom, bond, and topology state of G under P .
3. **Exact assembly verification.** A reverse transform decomposes G into role-assigned precursors, and the corresponding forward transform must reconstruct G exactly.
4. **Recursive synthesis-program construction.** FORGE searches for preparation routes to generated precursors that are not already supported as available starting materials and abstains when a route cannot be completed.

Figure 1 summarizes how FORGE separates reaction semantics from molecular identity and connects whole-graph generation to recursive synthesis-program construction.

3.2 SYNTHESIS-GROUNDED DESIGN TASK

Given a reaction program P , FORGE must generate a complete molecular graph G and either construct a complete synthesis program for G or abstain. Let $\mathfrak{T}$ denote the space of recursive synthesis programs and $\perp$ an explicit abstention outcome. We write the design process as

$$G \sim p_\theta(\cdot | P), \quad T = \mathcal{A}(G, P) \in \mathfrak{T} \cup \{\perp\}. \quad (6)$$

$\mathcal{A}$ denotes the bounded program constructor that verifies the final assembly, searches for preparation routes to unavailable precursors, and checks terminal supplier evidence. P and $\mathcal{A}$ act at different stages: P guides the generation of G , whereas $\mathcal{A}$ constructs T from the completed graph without modifying it. FORGE accepts (G, T) only when G passes molecular and exact-assembly verification and T traces every required precursor to an available starting material; otherwise, it abstains.

3.3 REACTION-PROGRAM SPECIFICATION

Definition 3.1 (Reaction program). A reaction program is a tuple

$$P = (c, d, \mathbf{r}, \mathbf{k}, M), \quad (7)$$

where c identifies the ordered assembly chemistry, d is the program depth, $\mathbf{r}$ assigns precursor roles, $\mathbf{k}$ marks reaction-core positions, and M specifies coarse per-role morphology. For each precursor role r , M records four integers: the number of exterior atoms n_r , junction budget j_r , cycle rank γ_r , and attachment count a_r .

M sets only coarse structural constraints; the model generates the exact topology, atoms, and bonds of each region. FORGE first samples M from a training-fold morphology prior and then generates G conditioned on the resulting reaction program:

$$p_{\theta}(G, M \mid c) = p_{\text{prog}}(M \mid c) p_{\theta}(G \mid M, c). \quad (8)$$

The prior sets each region’s atom count, branching budget, cycle rank, and number of core attachments, but not its molecular structure. A chemistry-specific adapter defines the precursor roles, reaction core, reverse decomposition, and forward transform associated with c . For each graph position i , FORGE combines learned embeddings of the program identity, program depth, precursor role, and reaction-core position:

$$h_i^P = \text{LN}(e_c(c) + e_d(d) + e_r(r_i) + e_k(k_i)). \quad (9)$$

These coordinates tell the generator which precursor role each position plays and whether it lies in the reaction core, but not which molecular structure should occupy that role.

3.4 PROGRAM-DEFINED MOLECULAR SPACE

Reaction programs constrain how a lipid is assembled without specifying the molecular structures that occupy its precursor roles. Let $\mathcal{G}_{\text{bounded}} \subseteq \mathcal{G}$ denote the molecular graphs within the declared atom and bond vocabularies, the 194-heavy-atom limit, and the three-closure limit. We write $G \models M(P)$ when its per-role atom counts, cycle ranks, and core-attachment counts match M , and its junction counts do not exceed the corresponding budgets. The valid, exact-L1 molecular space associated with P is

$$\mathcal{G}_{\text{L1}}(P) = \{G \in \mathcal{G}_{\text{bounded}} : \text{Valid}(G) = 1, G \models M(P), \text{L1}(G, P) = 1\}. \quad (10)$$

A generated lipid can pass the exact L1 check even when one or more recovered precursors never appeared in the training component set, provided the specified reaction reconstructs the lipid from them. For a reaction program P , let $\mathfrak{T}_P$ denote its admissible program inputs, including precursor identities, role assignments, and program depth. The registered forward operator maps each input to a finite set of canonical products:

$$\mathcal{F}_P : \mathfrak{T}_P \longrightarrow 2^{\mathfrak{G}_{N,K}}. \quad (11)$$

Given role-specific catalogues $\mathcal{C}$, let $\mathfrak{T}_P(\mathcal{C}) \subseteq \mathfrak{T}_P$ contain the admissible program inputs whose components all belong to their corresponding catalogues. The catalogue-reachable product set is

$$\mathcal{G}_{\text{cat}}(P, \mathcal{C}) = \bigcup_{\mathbf{C} \in \mathfrak{T}_P(\mathcal{C})} \mathcal{F}_P(\mathbf{C}). \quad (12)$$

For any graph G , define its complete program-relative preimage as

$$\mathcal{E}_P^*(G) = \{\mathbf{C} \in \mathfrak{T}_P : G \in \mathcal{F}_P(\mathbf{C})\}. \quad (13)$$

Definition 3.2 (Strong catalogue escape).

$$\text{Escape}^*(G; P, \mathcal{C}) = \mathbb{I}[\mathcal{E}_P^*(G) \neq \emptyset \wedge \mathcal{E}_P^*(G) \cap \mathfrak{T}_P(\mathcal{C}) = \emptyset]. \quad (14)$$

Theorem B.3 bounds the catalogue-reachable product set while establishing a zero component-novelty ceiling for the catalogue assembler. Strong catalogue escape additionally requires that no admissible all-catalogue input produces G . The verifier-returned set $\hat{\mathcal{E}}_P(G)$ defined below is a subset of $\mathcal{E}_P^*(G)$ when every returned trace passes exact forward replay. A novel component in a returned trace establishes strong catalogue escape only when reverse enumeration is exhaustive or direct catalogue reachability is checked independently. FORGE instead parameterizes $p_{\theta}(G \mid P)$ over whole graphs in $\mathcal{G}_{\text{bounded}}$, without including any component catalogue $\mathcal{C}_r$ in the generative state. Reaction programs constrain assembly semantics, while the flow determines precursor identity at atom and bond resolution.

3.5 WHOLE-GRAPH GENERATION

FORGE realizes this whole-graph distribution with a discrete flow over atom, bond, and topology states. To preserve the full 194-atom support without an $O(N_{\max}^2)$ edge state, FORGE encodes each graph as a spanning tree plus at most three residual closure edges. Let

$$X_G = (\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}) \quad (15)$$

denote the categorical graph state, where $\mathbf{a}$ contains atom states, $\boldsymbol{\pi}$ parent pointers, $\mathbf{b}$ spanning-tree bond states, $\mathbf{e}^L$ and $\mathbf{e}^R$ closure-endpoint pointers, and $\mathbf{z}$ closure-bond states. The sparse decoder is surjective onto the declared graph space $\mathfrak{G}_{N_{\max}, K}$: every graph in this space has at least one exact tree-closure encoding (Theorem B.2). For this study, $N_{\max} = 194$ and $K = 3$. This is a statement about architectural support. It does not imply reaction consistency, nonzero probability under a trained model, or synthesis-program completeness.

The chemistry adapter marks any atom, pointer, or bond states fixed by P . FORGE holds these states constant while corrupting the remaining states along the linear probability path and predicting their clean values from (X_t, t, P) . Let $\mathcal{Q} = \{\mathbf{a}, \boldsymbol{\pi}, \mathbf{b}, \mathbf{e}^L, \mathbf{e}^R, \mathbf{z}\}$ denote the six state families optimized during training. Let μ_{bal} denote the frozen training distribution over graph-program pairs, which assigns equal total mass to each reaction program and preserves the source-balanced weights within each program. For each $q \in \mathcal{Q}$, let $m_j^{(q)}(P) \in \{0, 1\}$ indicate whether coordinate j remains variable under P . FORGE minimizes a program-balanced denoising objective over the variable coordinates:

$$\mathcal{L}_{\text{FORGE}}(\theta) = \mathbb{E}_{(G, P) \sim \mu_{\text{bal}}, t \sim \mathcal{U}(0, 1), X_t \sim p_t(\cdot | X_G, P)} \left[\sum_{q \in \mathcal{Q}} \lambda_q \frac{\sum_j m_j^{(q)}(P) \left[-\log p_\theta(X_{G,j}^{(q)} | X_t, t, P) \right]}{\max\{1, \sum_j m_j^{(q)}(P)\}} \right]. \quad (16)$$

Here, λ_q weights state family q , and a family with no variable coordinates contributes zero loss. The corruption and sampling operators project every intermediate state onto the program-fixed coordinates. This makes those coordinates invariant by construction, while the masked denoising objective learns the conditional clean-state distributions of the remaining coordinates in the population limit (Proposition B.4 and Theorem B.5).

The denoiser is a graph Transformer over the complete sparse state. Each layer applies graph self-attention with relation biases for parent, child and closure edges, then cross-attends to memory tokens representing program identity, program depth, precursor role and reaction-core position. Cross-attention is repeated in every layer so that reaction semantics can influence both local state updates and long-range coordination among precursor-derived regions. Three softly program-routed residual adapters provide family-specific capacity while retaining a shared whole-graph backbone.

The complete training loss augments the flow objective with precursor-role and reaction-core consistency terms:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{FORGE}} + \lambda_{\text{role}} \mathcal{L}_{\text{role}} + \lambda_{\text{core}} \mathcal{L}_{\text{core}}. \quad (17)$$

Present semantic states receive equal mass within each auxiliary term. Training gives every reaction family equal total loss mass and uses deterministic projected conflicting gradient updates (PCGrad) to remove pairwise gradient conflicts without amplifying a nearly converged family’s gradient. Table 4 prespecifies separate interventions for layerwise cross-attention, each auxiliary loss, routed adapters and gradient conflict control.

At inference, FORGE samples X_0 from the source distribution, holds the program-fixed coordinates constant, and simulates $R_t^g(\cdot | P)$ from $t = 0$ to $t = 1$. FORGE decodes the terminal state X_1 into a molecular graph G and retains it only if it is connected, valence-valid, and passes molecular sanitization.

3.6 EXACT ASSEMBLY VERIFICATION

To ensure that each retained graph satisfies the specified reaction program, FORGE decomposes it into role-assigned precursors and applies the corresponding forward transform.

Definition 3.3 (Exact L1 consistency). For chemistry c , the reverse transform $\mathcal{D}_c$ maps G to a set of role-assigned precursor tuples. Let $\mathcal{D}_P(G) \subseteq \mathcal{D}_c(G)$ contain only those precursor tuples whose role assignments, reaction-core atoms, and morphology satisfy P . The verifier-returned exact set is

$$\widehat{\mathcal{E}}_P(G) = \{\mathbf{C} \in \mathcal{D}_P(G) : G \in \mathcal{F}_P(\mathbf{C})\}. \quad (18)$$

A graph G satisfies exact L1 consistency under P when this set is nonempty:

$$\text{L1}(G, P) = \mathbb{K}[\widehat{\mathcal{E}}_P(G) \neq \emptyset]. \quad (19)$$

Post-hoc filtering changes which completed samples are retained but cannot change the proposal distribution’s exact-L1 acceptance probability. Under independent sampling, this probability determines the complete distribution of accepted counts under a fixed attempt budget and the expected cost of obtaining a fixed number of accepted products (Proposition B.6). FORGE reports decomposition coverage separately from forward-replay precision because identifying a candidate disconnection does not guarantee exact reconstruction of G . Decomposition coverage is the fraction of valid graphs for which $\mathcal{D}_c(G) \neq \emptyset$; forward-replay precision is the fraction of returned precursor tuples for which $\mathcal{F}_c(\mathbf{C}) \cong G$. FORGE rejects G when $\mathcal{D}_c(G) = \emptyset$ and marks it as ambiguous when $|\mathcal{D}_c(G)| > 1$.

3.7 RECURSIVE SYNTHESIS-PROGRAM CONSTRUCTION

The final assembly requires access to every recovered precursor. Exact L1 verification identifies these precursors, but it does not establish whether they are available or how to prepare them. For laboratory planning, an exact final assembly is useful only when every required precursor is either available or connected to available starting materials through a supported preparation route. Let $T = (V_M, V_R, E)$ denote a bipartite synthesis tree rooted at G , where each product molecule points to its preparation reaction and each reaction points to its required reactants. FORGE constructs these routes recursively and rejects the lipid when any branch remains unresolved. A molecule node terminates when a current, dated supplier record lists an exact constitutional match as available; otherwise, FORGE must expand that node through a supported, forward-verified reaction. FORGE uses a bounded level-two (L2) planner to propose each expansion. For each proposal, FORGE records the reaction, reactants, product, and supporting evidence, and accepts the step only when the forward transform reconstructs the target precursor exactly. Each accepted reaction adds its reactants as new molecule nodes, and FORGE applies the same termination-or-expansion rule recursively. For each L2 reaction node r , let $\text{Verify}(r) = 1$ when its forward transform reconstructs its product exactly. For each leaf molecule ℓ , let $\text{Available}(\ell) = 1$ when it satisfies the terminal supplier-evidence rule. Let $V_R^{L2}(T)$ denote the L2 reaction nodes and $L(T)$ the leaf molecules in T .

$$\text{Complete}(G, T, P) = \text{L1}(G, P) \prod_{r \in V_R^{L2}(T)} \text{Verify}(r) \prod_{\ell \in L(T)} \text{Available}(\ell). \quad (20)$$

Let $\mathcal{B}$ fix the route-depth and query budgets, registered transforms, evidence-admission policy, and dated supplier snapshot used by the assessor $\mathcal{A}_{\mathcal{B}}$. Under finite search bounds and terminating evidence queries, the assessor terminates and returns a dossier only when $\text{Complete}(G, T, P) = 1$ (Lemma B.7 and Theorem B.8). An abstention means that FORGE did not complete a dossier under $\mathcal{B}$; it does not imply that no synthesis route exists outside the frozen search space or evidence snapshot. The guarantee is computational and does not establish experimental synthesis success.

Algorithm 1 summarizes the complete inference procedure. Every iteration consumes one generation attempt, and FORGE retains the outcome of every failed attempt in a typed ledger.

For every accepted pair (G, T) , FORGE records a computational synthesis dossier containing all molecules and reactions in T , the evidence supporting each reaction, and the dated supplier record for every terminal material. We classify each generated lipid into one of four support tiers according to which elements of its assembly lie outside the fixed reference space: **E0 (enumerated)**: the generated lipid matches a product that was already constructed computationally from known components using a supported reaction. **E1 (known components)**: the generated lipid was not included in the bounded E0 enumeration, but a supported reaction assembles it entirely from known components. **E2 (generated precursor)**: a supported reaction assembles the lipid, but at least one precursor is

Algorithm 1 FORGE generation and fail-closed synthesis-program construction.

Require: trained rates R_t^θ , reaction program P , transition count K , attempt budget B , recorded seed s , and frozen route contract $\mathcal{B}$

Ensure: accepted computational dossiers $\mathcal{S}$ and typed failure ledger $\mathcal{H}$

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1: initialize the random-number streams from  $s$ ;  $\mathcal{S} \leftarrow \emptyset$ ;  $\mathcal{H} \leftarrow \emptyset$ 
2: for  $b = 1, \dots, B$  do
3:   sample  $X_0 \sim p_0(\cdot | P)$  and set  $X \leftarrow \Pi_P(X_0)$ 
4:   for  $k = 0, \dots, K - 1$  do
5:      $t_k \leftarrow k/K$ 
6:      $X \leftarrow \Pi_P(\text{JUMPSTEP}(X, R_{t_k}^\theta(\cdot | X, P)))$ 
7:   end for
8:    $G \leftarrow \text{DECODE}(X)$ 
9:   if  $G$  is disconnected, valence-invalid, or fails molecular sanitization then
10:     $\mathcal{H} \leftarrow \mathcal{H} \cup \{(b, \text{INVALIDGRAPH})\}$ 
11:    continue
12:   end if
13:    $\hat{\mathcal{E}}_P(G) \leftarrow \{\mathbf{C} \in \mathcal{D}_P(G) : G \in \mathcal{F}_P(\mathbf{C})\}$ 
14:   if  $\hat{\mathcal{E}}_P(G) = \emptyset$  then
15:     $\mathcal{H} \leftarrow \mathcal{H} \cup \{(b, \text{L1FAILURE})\}$ 
16:    continue
17:   end if
18:    $Y \leftarrow \mathcal{A}_B(G, P)$  ▷ recursively assess the exact L1 precursors
19:   if  $Y$  is a tree  $T$  and  $\text{Complete}(G, T, P) = 1$  then
20:     $\mathcal{S} \leftarrow \mathcal{S} \cup \{\text{DOSSIER}(G, P, T)\}$ 
21:   else
22:     $\mathcal{H} \leftarrow \mathcal{H} \cup \{(b, Y)\}$  ▷ preserve the typed outcome
23:   end if
24: end for
25: return  $\mathcal{S}, \mathcal{H}$ 

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newly generated and requires a complete L2 route to available starting materials. **E3 (new assembly family):** the lipid requires a final assembly reaction outside the qualified reaction registry.

4 RESULTS

4.1 EXPERIMENTAL SETTINGS

Our evaluation spans the three assembly programs defined above, with Ugi as the deeply supervised case, repeated aza-Michael addition as a second family, and repeated reductive amination as a lighter stress test. The evaluation tests four capabilities: learning multiple assembly programs within one model, using reaction semantics during generation rather than after it, producing exact assemblies beyond a finite component catalogue, and tracing generated precursors to available starting materials. We compare four matched training arms: Ugi-only conditioning, shared conditioning across all three programs, shared training without program information, and shared training with cyclically incorrect program labels. All arms use the same architecture and optimization budget, and we evaluate the prespecified final checkpoint from each of three independent seeds. Each matched comparison uses the same number of generation attempts per program. Our primary metric is exact-L1 yield per generation attempt: the number of generated graphs that pass molecular validity, program-aware decomposition, and exact forward replay divided by the total number of sampling attempts. Secondary metrics separate graph quality (validity and connectedness), reaction recovery (coverage, precision, abstention, and ambiguity), and exploration (unique-product fraction, whole-product and component novelty, molecular diversity, and effective component count). We report means across seeds, individual seed values, and 95% paired-seed bootstrap intervals; generated molecules are not treated as independent replicates. Training uses only the component-disjoint training folds. Calibration data are diagnostic, and held-out products neither enter training nor select checkpoints. The three shared arms assign equal total sampling mass to each program while preserving the frozen within-program

source weights. We assess whether shared training preserves Ugi behavior using prespecified non-inferiority margins and require zero program-fixed-state violations and zero support overflows.

Two additional comparisons isolate the role of reaction semantics. First, we compare conditioned generation with null generation followed by exact post-hoc reaction filtering under the same attempt budget; the cyclic-label arm tests whether chemical meaning matters beyond the presence of a categorical input. Second, we compare FORGE with a train-only finite-component assembler that receives the exact training components, supported forward chemistry, source-weighted role marginals, and the same number of attempts. We report reaction validity and exploration separately because the catalogue can achieve high exact-L1 yield while having zero input-component novelty by construction. These comparisons make no L2 planner, biological-oracle, or candidate-selection calls.

For the Ugi program, we additionally define a common external-baseline protocol with one reaction-space comparator, RGFN, and two generic whole-molecule comparators, unconditional DeFoG and GenMol/SAFE (Kozlowski et al., 2024; Qin et al., 2025; Lee et al., 2025). The Synthesis-DAG method of Ou et al. (2024), SynFlowNet and SynCoGen remain methodological context rather than empirical rows (Cretu et al., 2025; Rekesh et al., 2025). The released Synthesis-DAG repository does not contain the paper-specific ionizable-lipid implementation, the released SynFlowNet action space cannot represent the qualified three-reactant Ugi transform, and the released SynCoGen repository has no usable code license. We do not replace missing functionality with an imitation, alter Ugi into sequential pseudo-reactions, or execute unlicensed code. An external row enters the numerical comparison only after the method has been executed on the same training split, generation-attempt budget, seed set, and common molecular and exact-L1 evaluation pipeline used for FORGE. Published values obtained on the method’s original dataset are not mixed with results from this common evaluation. Methods that do not support the additional assembly programs are compared on Ugi only rather than assigned artificial aza-Michael or reductive-amination results. The matched internal suite separately includes unconditioned generation plus post-hoc filtering, an oracle finite-catalogue assembler, a learned inventory selector, two learned component-factorized generators, and targeted interventions on the Transformer. Table 4 reports the completed three-seed Transformer mechanism and component-factorization study. Each intervention uses the same folds, generation budget and evaluation code as the full Transformer.

4.2 SHARED REACTION-PROGRAM LEARNING ACROSS THREE ASSEMBLY CHEMISTRIES

Across 3,072 held-out generation attempts per program in each of three independent training seeds, the final conditioned model achieved exact-L1 yields of $78.7 \pm 10.1\%$ for Ugi (72.3%, 90.3% and 73.5% by seed), $77.5 \pm 3.7\%$ for repeated aza-Michael addition (76.5%, 81.5% and 74.3%), and $31.3 \pm 14.3\%$ for repeated reductive amination (38.6%, 14.8% and 40.4%) (mean $\pm$ sample standard deviation; Figure 2 and Table 1). Exact forward-replay precision was 100% in every program and seed. However, mean decomposition ambiguity was 54.9% for repeated aza-Michael addition and 74.1% for repeated reductive amination, and the latter program showed substantial between-seed variation. Table 1 reports molecular validity, connectedness, decomposition coverage, forward-replay precision, abstention, ambiguity, novelty and diversity separately for each program.

Relative to the Ugi-only model, shared training changed Ugi exact-L1 yield by 53.26 percentage points (95% paired-seed interval 41.24 to 61.82) and met every prespecified Ugi-retention gate. The final conditioned model produced zero fixed-state violations and zero support overflows across 27,648 held-out attempts (three seeds, three programs and 3,072 attempts per program). Together, these results support the conclusion that one conditional flow can learn the three specified reaction-program coordinates without erasing the deeply supported Ugi behavior.

4.3 REACTION SEMANTICS ALTER GENERATION RATHER THAN MERELY FILTERING ITS OUTPUT

Under matched attempt budgets, conditioned generation produced 786.8, 774.6 and 312.7 exact-L1 products per 1,000 attempts for Ugi, repeated aza-Michael and repeated reductive amination, compared with 114.4, 39.7 and 131.7 for matched null generation followed by post-hoc filtering (Figure 4a–c). The paired conditioned-minus-post-hoc differences were 67.24 percentage points for

Exact-L1 yield (%) and transform verification across three seeds				
Program	FORGE conditioned	Shared null	Cyclic program	Decomp./replay (%)
Ugi	78.7 $\pm$ 10.1	11.4 $\pm$ 1.1	40.2 $\pm$ 0.7	90.7/100.0
Aza-Michael	77.5 $\pm$ 3.7	4.0 $\pm$ 0.8	3.8 $\pm$ 0.6	100.0/100.0
Reductive amination	31.3 $\pm$ 14.3	13.2 $\pm$ 1.3	18.1 $\pm$ 0.8	84.3/100.0

Shared conditioning improved Ugi exact-L1 yield over Ugi-only training by 53.26 percentage points (95% interval 41.24 to 61.82).

Figure 2: **A shared model learns multiple ionizable-lipid assembly programs.** Results are reported separately for Ugi, repeated aza-Michael addition and repeated reductive amination. Cells show means and sample standard deviations across three independent training seeds. The stated Ugi contrast uses a 95% paired-seed bootstrap interval. The Ugi-only arm is evaluated only on Ugi, and results are not pooled across reaction programs. All values in this figure and Table 1 are hash verified.

Table 1: **Matched production comparison across reaction programs.** Point estimates report the mean across three independent seeds; the exact-L1 column additionally reports the sample standard deviation. Paired contrasts include 95% paired-seed bootstrap intervals; individual seed values appear in the supplement. Validity and connectedness use all generation attempts as the denominator. Decomposition coverage, replay precision, abstention and ambiguity retain their prespecified conditional denominators. Novelty is conditioned on valid products or unique returned decompositions, as appropriate. Fraction pairs and exact-L1 yield are percentages; diversity is mean pairwise ECFP4 distance and N_{eff} is the effective generated-component count. Results are never pooled across reaction programs.

Program	Arm	Valid./Conn.	Decomp./Replay	Abst./Ambig.	Exact-L1/attempt	Whole/Comp. novelty	Diversity/ N_{eff}
Ugi	Ugi-only	28.0/28.0	89.9/100.0	10.1/0.0	25.4 $\pm$ 12.0	99.8/92.4	0.597/134.8
Ugi	FORGE conditioned	86.6/86.6	90.7/100.0	9.3/0.0	78.7 $\pm$ 10.1	99.8/96.2	0.621/320.8
Ugi	Shared null	19.5/19.5	58.8/100.0	41.2/0.0	11.4 $\pm$ 1.1	100.0/99.9	0.754/219.9
Ugi	Cyclic program	43.4/43.4	92.7/100.0	7.3/0.0	40.2 $\pm$ 0.7	99.9/90.1	0.562/138.0
Aza-Michael	FORGE conditioned	77.5/77.5	100.0/100.0	0.0/54.9	77.5 $\pm$ 3.7	99.7/99.2	0.521/156.9
Aza-Michael	Shared null	16.5/16.5	24.3/100.0	75.7/0.1	4.0 $\pm$ 0.8	99.9/99.7	0.635/111.3
Aza-Michael	Cyclic program	25.7/25.7	15.2/100.0	84.8/0.0	3.8 $\pm$ 0.6	99.8/98.3	0.388/51.5
Reductive amination	FORGE conditioned	36.5/36.5	84.3/100.0	15.7/74.1	31.3 $\pm$ 14.3	98.1/100.0	0.624/102.9
Reductive amination	Shared null	15.4/15.4	85.4/100.0	14.6/70.7	13.2 $\pm$ 1.3	99.5/100.0	0.615/77.2
Reductive amination	Cyclic program	21.4/21.4	84.9/100.0	15.1/74.6	18.1 $\pm$ 0.8	96.2/100.0	0.570/34.8

Ugi (95% interval 60.84 to 79.95), 73.49 for aza-Michael (69.69 to 78.45), and 18.10 for reductive amination (2.51 to 26.04).

The cyclic control retained precursor roles, reaction-core coordinates and program depth but assigned the wrong reaction identity. Relative to correct conditioning, this intervention changed exact-L1 yield by 38.43 percentage points for Ugi (95% interval 32.00 to 49.41), 73.67 for aza-Michael (71.13 to 77.02), and 13.15 for reductive amination (-3.91 to 22.01). Forward-replay precision remained 100% for every returned decomposition in these arms.

A separate inference-time intervention held the trained network and Ugi request fixed while changing only program coordinates. The factual exact-L1 yield was 39.69%. Nulling all program coordinates reduced it to 3.62%, a 36.07-point loss (95% interval 34.28 to 37.24). Reversing the role assignment caused a 4.49-point loss (1.37 to 8.27), and substituting the reductive-amination program token caused a 22.60-point loss (14.06 to 27.47). The aza-Michael-token contrast was unresolved at 2.00 points (-2.18 to 5.60). Thus the coordinates affect generation, but the evidence does not support an equal or individually identifiable effect for every coordinate.

4.4 WHOLE-GRAPH GENERATION PRODUCES PRECURSOR IDENTITIES BEYOND A FINITE VOCABULARY

Table 2 defines the common Ugi benchmark used to compare FORGE with external synthesis-aware generators, matched internal controls, and the finite-catalogue oracle. Every yield is normalized by the number of generation attempts, including attempts that fail before producing a valid molecule. All included methods were run under the common split, attempt budget, seed set and verifier.

Table 2: **Common Ugi benchmark for external generators and matched controls.** All yields are reported per 1,000 generation attempts under the same component-disjoint split, three-seed protocol, and common molecular and exact-L1 verifier. Numerical entries are mean $\pm$ sample standard deviation across the three independent seeds; the seed-level values are reported in Table 9. An open-ended exact-L1 product has exactly one verifier-returned decomposition and at least one component constitution absent from the method-visible training catalogue. The dagger marks a zero imposed by a method’s finite component action space, not an empirical failure. Published values from the original baseline papers are not inserted into this table. N/E denotes a metric that cannot be estimated because the method produced no valid molecular output.

Method	Valid/1k	Exact L1/1k	Unique L1/1k	Open-ended/1k	Held-comp. L1/1k	Diversity
<i>Reaction-space external baselines</i>						
RGFN	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0 [†]	0.0 $\pm$ 0.0	N/E
<i>Generic whole-molecule external baselines with common post-hoc assessment</i>						
DeFoG unconditional	346.5 $\pm$ 42.3	280.6 $\pm$ 26.6	280.6 $\pm$ 26.6	280.6 $\pm$ 26.6	15.5 $\pm$ 2.4	0.727 $\pm$ 0.014
GenMol/SAFE	627.5 $\pm$ 6.2	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.873 $\pm$ 0.009
<i>Matched controls and formulation baselines</i>						
Learned inventory selector	1000.0 $\pm$ 0.0	1000.0 $\pm$ 0.0	849.5 $\pm$ 13.7	0.0 $\pm$ 0.0 [†]	0.0 $\pm$ 0.0	0.717 $\pm$ 0.001
Finite catalogue oracle	1000.0 $\pm$ 0.0	1000.0 $\pm$ 0.0	837.6 $\pm$ 8.2	0.0 $\pm$ 0.0 [†]	0.0 $\pm$ 0.0	0.719 $\pm$ 0.000
Shared-null + post-hoc	194.7 $\pm$ 15.6	114.4 $\pm$ 10.9	108.5 $\pm$ 9.7	108.5 $\pm$ 9.7	0.4 $\pm$ 0.2	0.754 $\pm$ 0.014
FACT-matched	60.8 $\pm$ 20.2	52.0 $\pm$ 15.6	50.8 $\pm$ 14.4	50.8 $\pm$ 14.4	0.0 $\pm$ 0.0	0.667 $\pm$ 0.009
FACT-generous	500.7 $\pm$ 213.8	296.7 $\pm$ 186.0	288.3 $\pm$ 175.3	288.3 $\pm$ 175.3	0.1 $\pm$ 0.2	0.774 $\pm$ 0.007
FORGE Transformer	866.0 $\pm$ 90.2	786.8 $\pm$ 100.8	751.2 $\pm$ 80.2	751.2 $\pm$ 80.2	0.1 $\pm$ 0.2	0.621 $\pm$ 0.011

Table 3: **Method-blind structural-realism diagnostic.** Every entry is mean $\pm$ sample standard deviation across three training seeds and retains all 3,072 requested attempts in the denominator. Support is the declared C/N/O/P/S and 194-heavy-atom space. FP and descriptor P/C report manifold precision per 1,000 attempts and coverage of the source-study-held-out R0 reference in percent. Desc. W is the normalized descriptor Wasserstein distance; lower is closer. C2ST is grouped classifier two-sample area under the receiver operating characteristic curve; 0.5 is indistinguishable. Unique is conditioned on connected outputs. N_{eff} is the Shannon effective molecule count. N/E denotes a metric that cannot be estimated. This post-hoc structural diagnostic is not a property, activity or synthesis-success score.

Method	Connected/1k	Support/1k	FP P/C	Desc. P/C	Desc. W	C2ST	Unique (%)	N_{eff}	Diversity
RGFN	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	N/E	N/E	N/E	0.0 $\pm$ 0.0	N/E
DeFoG unconditional	346.5 $\pm$ 42.3	334.2 $\pm$ 39.1	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	17.3 $\pm$ 1.4/1.45 $\pm$ 0.06	2.067 $\pm$ 0.009	0.961 $\pm$ 0.007	100.0 $\pm$ 0.0	1064.3 $\pm$ 129.9	0.726 $\pm$ 0.013
GenMol/SAFE	627.5 $\pm$ 6.2	627.4 $\pm$ 6.2	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	3.199 $\pm$ 0.003	N/E	0.9 $\pm$ 0.1	3.2 $\pm$ 0.1	0.873 $\pm$ 0.009
Finite catalogue oracle	1000.0 $\pm$ 0.0	956.5 $\pm$ 2.2	0.2 $\pm$ 0.2/0.05 $\pm$ 0.05	54.6 $\pm$ 4.4/2.02 $\pm$ 0.17	2.107 $\pm$ 0.002	0.971 $\pm$ 0.007	83.8 $\pm$ 0.8	2395.5 $\pm$ 31.7	0.720 $\pm$ 0.000
Learned inventory selector	1000.0 $\pm$ 0.0	945.0 $\pm$ 6.5	0.1 $\pm$ 0.2/0.02 $\pm$ 0.03	56.6 $\pm$ 7.2/1.97 $\pm$ 0.46	2.101 $\pm$ 0.005	0.974 $\pm$ 0.002	85.0 $\pm$ 1.4	2443.0 $\pm$ 55.8	0.718 $\pm$ 0.001
FORGE Transformer	866.0 $\pm$ 90.2	864.6 $\pm$ 91.2	0.0 $\pm$ 0.0/0.00 $\pm$ 0.00	8.8 $\pm$ 2.3/1.16 $\pm$ 0.20	2.089 $\pm$ 0.005	0.973 $\pm$ 0.019	95.7 $\pm$ 1.9	2479.4 $\pm$ 177.2	0.621 $\pm$ 0.013

The learned finite-inventory selector achieved 1000.0 ± 0.0 exact-L1 products per 1,000 attempts, but its finite action space imposed zero open-ended component generation. FORGE produced 786.8 ± 100.8 exact-L1 products and 751.2 ± 80.2 unique open-ended exact-L1 products per 1,000 attempts. Among open-ended external generators, unconditional DeFoG produced 280.6 ± 26.6 exact-L1 products per 1,000 attempts and recovered held components at 15.5 ± 2.4 per 1,000 attempts, compared with 0.1 ± 0.2 for FORGE. GenMol/SAFE produced 627.5 ± 6.2 valid connected outputs but no exact Ugi-L1 product, while RGFN emitted no admissible molecular output under the strict native contract. These outcomes expose separate tradeoffs between reaction consistency, open-ended identity generation and recovery of specifically reserved components.

No evaluated method reproduced the full held-out observed-lipid distribution: where estimable, grouped C2ST AUC remained between 0.961 and 0.974, and fingerprint-manifold precision was zero or near zero (Table 3). GenMol/SAFE’s high pairwise diversity was paired with an effective count of only 3.2 ± 0.1 molecules, confirming severe frequency collapse. FORGE retained 2479.4 ± 177.2 effective molecules, but only 8.8 ± 2.3 attempts per 1,000 entered the descriptor manifold. We therefore interpret this analysis as a multi-axis structural diagnostic rather than a single measure of whether a generated lipid is reasonable.

4.5 JOINT WHOLE-LIPID DENOISING HAS REACTION-DEPENDENT ADVANTAGES OVER ROLE ISOLATION

The full Transformer produced 791.8 Ugi and 762.5 repeated aza-Michael exact-L1 products per 1,000 attempts, compared with 52.0 and 393.7 for the parameter-matched role-isolated FACT model (Table 4). The full Transformer and FACT-matched model each had 5.33 million trainable parameters; FACT-generous had 29.40 million. The higher-capacity FACT control improved to 296.7 for

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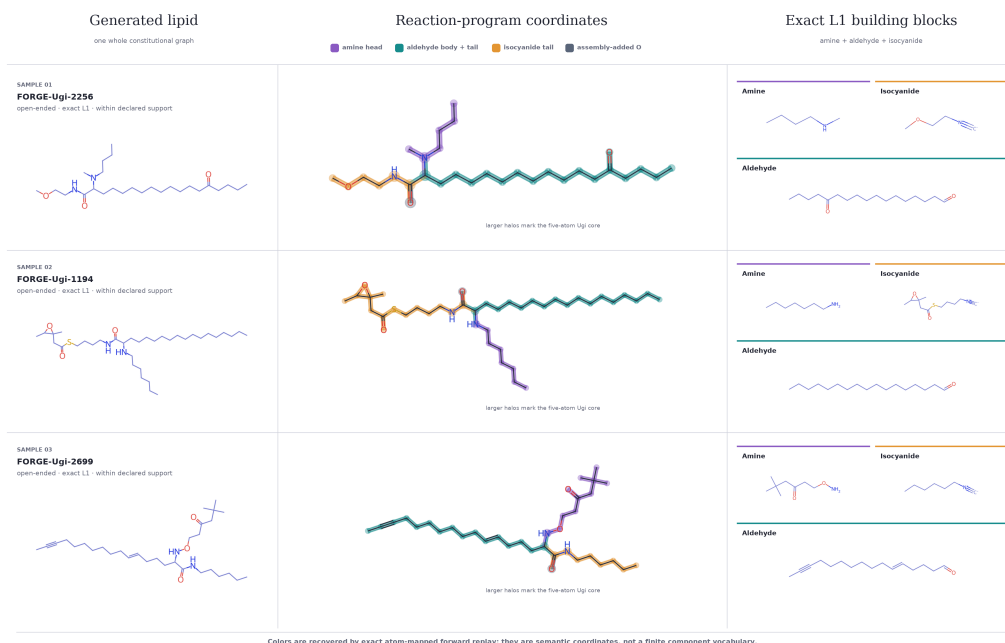


Figure 3: **Deterministic examples of open-ended FORGE ionizable-lipid graphs.** Each row is one seed-0 Ugi product with a unique exact-L1 decomposition, at least one recovered precursor constitution outside the method-visible training inventory, membership in the frozen descriptor manifold and the declared molecular support. The three displayed identities are the deterministic SHA-256-seeded ECFP4 MaxMin display set among 23 eligible products; they were not selected as experimental candidates. Left, the complete constitutional graph sampled jointly by FORGE. Center, the same graph annotated by precursor origin recovered through exact atom-mapped forward replay; larger halos mark the five-atom Ugi core. Right, the exact amine, aldehyde and isocyanide precursor constitutions returned by the common Ugi verifier. These colors are semantic coordinates, not a finite component vocabulary from which the model selects. Descriptor-manifold membership and exact L1 replay are structural diagnostics, not evidence of activity or synthesis success.

Ugi and 687.0 for repeated aza-Michael addition, but remained below the full Transformer in every seed for both programs. The direction reversed for repeated reductive amination: the full Transformer produced 318.1 exact-L1 products per 1,000 attempts, whereas FACT-matched and FACT-generous produced 509.9 and 561.3. Thus, joint whole-lipid denoising improved exact assembly for Ugi and repeated aza-Michael addition but was not uniformly superior across reaction programs. Because the prespecified conditional-dependence diagnostic was negative, this architecture contrast does not establish detectable cross-role dependence in the present data.

Removing layerwise program cross-attention reduced mean exact-L1 yield by 54.5 products per 1,000 attempts for Ugi and 40.8 for repeated reductive amination, while increasing repeated aza-Michael yield by 17.4. Removing the role or core objective and removing routed adapters also produced program-dependent changes rather than a uniform loss. The arm without gradient-conflict control slightly exceeded the full Transformer mean in all three programs. We therefore do not claim that every introduced module independently improves exact-L1 yield. These one-at-a-time interventions identify which effects replicate across chemistries; they are not pooled into a composite architecture score.

At the matched sampling budget, FORGE produced 751.2 unique method-visible open-ended exact-L1 Ugi products per 1,000 attempts, averaged across the three independent seeds. The corresponding values were 313.9 for aza-Michael addition and 34.8 for reductive amination. The finite-component assembler produced zero by construction because every sampled product uses only components from the training catalogue (Figure 4d-f and Table 5). The catalogue achieved exact-L1 yields of 1,000.0, 723.2 and 1,000.0 products per 1,000 attempts, compared with 786.8, 774.6 and 312.7 for

Table 4: **Transformer architecture and mechanism ablations.** The five Transformer intervention rows change only the named mechanism from the full model; the two FACT rows instead generate precursor roles independently at matched or increased capacity. All arms use identical data folds, corruption, context schedule, checkpoint rule and sampling budget. Entries report the mean and [95% seed-bootstrap interval] across three independent training seeds. Held-out loss is the mean denoising loss at $t = 0.5$ across the three component-disjoint program partitions. Cross-role fidelity is the Frobenius distance to the held-out residualized descriptor matrix, for which lower values indicate closer agreement. Exact-L1 yields use all 3,072 attempts per program and seed as the denominator.

Arm	Held-out loss	Ugi L1/1k	BL L1/1k	LX L1/1k	Held-comp. L1/1k	Cross-role fidelity	Diversity
Input-only program	8.48 [7.72, 9.14]	737.30 [631.18, 845.70]	779.84 [757.49, 820.31]	277.34 [230.14, 339.52]	0.11 [0.00, 0.33]	1.79 [1.70, 1.91]	0.60 [0.59, 0.61]
No role loss	7.87 [7.78, 8.00]	771.59 [672.20, 889.00]	772.14 [743.49, 820.64]	289.28 [115.23, 396.16]	0.11 [0.00, 0.33]	1.94 [1.89, 1.98]	0.59 [0.58, 0.60]
No core loss	7.85 [7.78, 7.98]	753.69 [674.15, 906.58]	772.79 [726.89, 819.34]	300.46 [151.69, 409.18]	0.11 [0.00, 0.33]	1.89 [1.75, 1.98]	0.59 [0.58, 0.59]
No routed adapters	7.81 [7.44, 8.04]	718.64 [592.45, 875.33]	779.08 [744.47, 823.89]	326.71 [265.62, 384.11]	0.00 [0.00, 0.00]	1.88 [1.75, 2.06]	0.59 [0.59, 0.59]
No gradient conflict control	7.98 [7.82, 8.20]	798.07 [732.42, 916.02]	768.88 [721.68, 806.32]	326.93 [149.09, 427.73]	0.11 [0.00, 0.33]	1.96 [1.87, 2.09]	0.59 [0.58, 0.59]
FACT-matched	7.09 [6.86, 7.34]	51.97 [37.76, 68.68]	393.66 [327.47, 460.94]	509.87 [457.36, 562.83]	0.00 [0.00, 0.00]	1.45 [1.14, 1.72]	0.56 [0.55, 0.57]
FACT-generous	8.24 [8.03, 8.60]	296.66 [161.78, 508.79]	686.96 [584.64, 811.52]	561.31 [491.54, 635.74]	0.11 [0.00, 0.33]	1.16 [1.14, 1.18]	0.60 [0.60, 0.60]
FORGE Transformer	7.88 [7.68, 8.12]	791.78 [725.91, 916.67]	762.48 [725.26, 815.10]	318.14 [134.44, 413.41]	0.11 [0.00, 0.33]	1.94 [1.76, 2.09]	0.59 [0.58, 0.59]

FORGE. The corresponding raw validities were 1,000.0, 723.2 and 1,000.0 for the catalogue and 866.0, 774.6 and 365.0 for FORGE. We report these quantities separately because reaction validity and verifier-relative component novelty measure different capabilities.

In a distinct-product Ugi role audit, 19,520 of 26,235 products (74.4%) contained at least one verifier-returned component constitution absent from the training registry. Across all Ugi products admitted to the novelty audit, 21,960 of 26,235 distinct products (83.7%) were absent from the full 112,386-product constitutional reference corpus. FORGE effective component counts were 320.8, 156.9 and 102.9, compared with 86.4, 23.4 and 16.9 for the finite catalogue. The Ugi catalogue contained 185 amines, 75 aldehydes and 35 isocyanides and covered 100% of training products but 0% of the calibration and component-disjoint held-out products. Its Cartesian tuple ceilings were 485,625 for Ugi, 774 for aza-Michael and 490 for reductive amination; these count possible component-program inputs rather than unique forward products. These results establish the intended tradeoff: the exact finite catalogue has high transform consistency, whereas whole-graph generation can produce exact-L1 products whose recovered precursor identities are not restricted to that inventory.

Exact assembly versus the finite-catalogue ceiling				
Program	FORGE L1/1k	Catalogue L1/1k	FORGE open-ended/1k	Catalogue open-ended/1k
Ugi	786.8	1000.0	729.3	0.0
Aza-Michael	774.6	723.2	313.9	0.0
Reductive amination	312.7	1000.0	34.8	0.0

Nulling all program coordinates reduced factual Ugi exact-L1 yield from 39.69% to 3.62%; substituting the reductive-amination token reduced it to 17.09%.

Figure 4: **Reaction conditioning improves sampling efficiency without imposing a finite component vocabulary.** The inset reports inference-time coordinate interventions, and the table compares FORGE with a train-only finite-component assembler that receives exact forward chemistry and the same per-program attempt budget. Values are means across three independent training seeds. The comparisons use identical molecular support and final reaction adapters and make no L2-planner, biological-oracle or candidate-selection calls. The finite catalogue has zero open-ended yield by construction.

Table 5: **Whole-graph generation versus a finite-component assembler.** Exact-L1, unique exact-L1 and open-ended exact-L1 yields are reported per 1,000 generation attempts. An operationally open-ended product has exactly one verifier-returned exact-L1 decomposition and contains at least one component constitution absent from the training catalogue. This verifier-relative metric is not called strong catalogue escape unless reverse enumeration is shown to be exhaustive. The catalogue has zero input-component novelty by construction, but may achieve higher raw validity or exact-L1 yield. FORGE-minus-catalogue contrasts use the three seeds as paired resampling units.

Program	Method	Valid/1k	Exact L1/1k	Unique L1/1k	Open-ended/1k	Whole nov.	Comp. nov.	Diversity/ N_{eff}
Ugi	FORGE	866.0	786.8	751.2	729.3	99.8	96.2	0.621/320.8
Ugi	Finite catalogue	1000.0	1000.0	837.6	0.0	16.9	0.0	0.719/86.4
Aza-Michael	FORGE	774.6	774.6	738.0	313.9	99.7	99.2	0.521/156.9
Aza-Michael	Finite catalogue	723.2	723.2	147.5	0.0	39.8	0.0	0.402/23.4
Reductive amination	FORGE	365.0	312.7	279.7	34.8	98.1	100.0	0.624/102.9
Reductive amination	Finite catalogue	1000.0	1000.0	52.1	0.0	44.8	0.0	0.514/16.9

4.6 NATIVE RECOVERY OF HELD COMPONENTS REMAINS WEAK

The component-disjoint Ugi fold was never used for training, checkpoint selection or inventory construction. Under the common native-generation assessment, the full Transformer recovered only 1 exact-L1 product containing a designated held component in 9,216 attempts, or 0.11 products per 1,000 attempts averaged across three seeds. This is a negative generalization result. It does not conflict with the broader verifier-relative novelty measurements above: generating a component absent from the training registry is easier than recovering one of the specifically reserved held-component identities. We therefore do not claim strong held-component recovery, role-specific held-component coverage, or zero-shot held-reaction-family generalization.

4.7 RECURSIVE SYNTHESIS PROGRAMS CONNECT GENERATED LIPIDS TO EVIDENCE-QUALIFIED MATERIALS

Table 6 separates a method’s native identity source and route representation from evidence obtained under the common FORGE assessor. We freeze the union of exact-L1 components from all methods and seeds before route assessment, hide method membership from the evidence pass, and require one complete, unresolved or search-censored disposition for every component before unsealing membership. Emitting a reaction sequence is not counted as a complete dossier unless the final assembly and every upstream step pass exact forward verification and every terminal material closes under the frozen evidence policy. We do not report a separate AiZynthFinder row: a proposal-only route is a hypothesis and cannot change exact evidence closure until every step passes the same forward-verification and terminal-evidence gates (Genheden et al., 2020).

The frozen union contained 11,313 distinct role–constitution components, but the exact-source library closed only 29. Consequently, FORGE produced 786.8 ± 100.8 exact-L1 products per 1,000 attempts, yet 0.0 ± 0.0 complete dossiers and 786.8 ± 100.8 abstentions under this library. The learned inventory selector closed 0.7 ± 0.3 dossiers per 1,000 attempts through occasional reuse of covered training components. This is a negative result about current exact evidence coverage, not a comparison of intrinsic synthesizability: components outside the frozen library were not searched and remain unknown.

The completed bounded route cascade starts from a frozen, route-blinded shortlist of 256 distinct Ugi products rather than from the full production-attempt population. All 256 products passed exact L1 reverification; 255 entered route assessment and one silicon-containing state was support-censored. Across the full shortlist, 19 closed under the exact-evidence and terminal-material policy, 236 remained unresolved and 1 was search-censored. The products contained 158 unique component targets: 36 complete, 120 unresolved and 2 search-censored. The proposal engines added graph-consistent hypotheses but closed no additional exact routes. Most unresolved components were absent from the bounded evidence index and were never expanded, so the result measures closure under this frozen cascade rather than molecular synthesizability.

Table 6: **Common synthesis-evidence assessment of Ugi generative methods.** Descriptive support fields record what each method is permitted to use at generation time. Numerical evidence fields are populated only after applying the same method-blind union of exact-L1 components, upstream forward-verification, terminal-evidence, and abstention policy to each method’s outputs. N/P means that the method does not produce that object; it is not interpreted as a measured zero. All numerical fields are reported per 1,000 generation attempts as mean $\pm$ sample standard deviation across three seeds. An abstention records missing exact evidence, not a failed synthesis; the frozen exact-source component library is deliberately non-exhaustive.

Method	Molecular identity source	Finite vocab.	Exact L1/Lk	Verified upstream	Terminal evidence	Complete dossier	Abstention
RGFN	reaction $\times$ building block	yes	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
DeFoG unconditional	generated atoms and bonds	no	280.6 $\pm$ 26.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	280.6 $\pm$ 26.6
GenMol/SAFE	generated fragment sequence	no	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Finite catalogue oracle	train components	yes	1000.0 $\pm$ 0.0	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	0.1 $\pm$ 0.2	999.9 $\pm$ 0.2
Learned inventory selector	selected train components	yes	1000.0 $\pm$ 0.0	0.7 $\pm$ 0.3	0.7 $\pm$ 0.3	0.7 $\pm$ 0.3	999.3 $\pm$ 0.3
Shared-null + post-hoc	generated atoms and bonds	no	114.4 $\pm$ 10.9	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	114.4 $\pm$ 10.9
FACT-matched	role-isolated denoising on shared count-only layout	no	52.0 $\pm$ 15.6	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	52.0 $\pm$ 15.6
FORGE Transformer	generated atoms and bonds	no	786.8 $\pm$ 100.8	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	786.8 $\pm$ 100.8

Bounded route-cascade dispositions				
Unit	Population	Complete	Unresolved	Censored
Products	256	19	236	1
Unique components	158	36	120	2

All 256 shortlist products passed exact-L1 reverification; one product was support-censored before route assessment. Proposal engines added hypotheses but no additional exact route closure.

Figure 5: **Recursive synthesis programs produce auditable computational dossiers.** Panel a reports the count and denominator at every stage; no stage is normalized to a later survivor set without stating that denominator. An L2 step is admitted only when its evidence tier permits the claimed reaction scope and its forward transform reconstructs the intended precursor. A terminal branch closes only with a current, dated supplier record for an exact constitutional match. Computational route closure establishes transform consistency and evidence-qualified terminal materials, not experimental synthesis success.

4.8 GUIDANCE DIAGNOSTICS DEFINE THE LIMITS OF ADDITIONAL OPTIMIZATION

At synthesis-guidance strength 0.25, the matched controller evaluated 64 terminal particles and retained 56 canonical representatives per arm. It changed only 8 terminal identities and returned 2 unique route-ready products, compared with 3 after post-hoc assessment. Only five of 48 four-particle morphology groups contained mixed route-readiness values, leaving most ancestry decisions uniform. This negative result does not support promoting mid-trajectory synthesis tilting. The production method therefore remains complete generation followed by proposal-augmented L1/L2/L3 route assessment.

We evaluate biological guidance only in the bounded HeLa messenger-transfection-potential (mTP) diagnostic. A group-disjoint preterminal classifier detected weak morphology-level signal, with ROC AUC 0.623 (95% interval 0.565 to 0.679) and a 1.177-fold observed top-quartile gain. We then compared a frozen applicability-enriched morphology proposal with a nested potency-enriched proposal using 3,072 generator calls per arm. Table 7 reports the matched terminal outcome.

The nested-minus-support difference was -1.30 conservative-high products per 1,000 generator calls (95% interval -2.28 to 0.98). The promotion gate failed. These computations evaluate predictor behavior within a declared applicability domain; they do not establish improved biological activity, delivery or experimental potency.

5 DISCUSSION

FORGE couples whole-graph generation to reaction programs and recursive synthesis programs. Reaction programs constrain precursor roles, reaction-core positions and transformation order with-

Table 7: Predicted HeLa messenger-transfection potential under matched generation budgets. The support-enriched arm uses the already promoted applicability proposal; the nested arm reallocates morphology mass using the frozen weak potency signal. Conservative-high counts are computational predictions, not transfection measurements. The comparison uses the same generator-call and oracle budgets.

Arm	Attempts	Exact L1	Eligible before budget	Oracle budget	Unique conservative-high
Support-enriched	3,072	2955	90	38	27
Nested potency proposal	3,072	2962	82	38	23

out specifying component identities. This allows the model to generate new molecular structures while exact forward replay and recursive routing keep accepted products connected to chemistry.

Across three assembly programs, the shared model achieved high exact-L1 yield for Ugi and repeated aza-Michael addition and lower, more variable yield for reductive amination. Matched null and cyclic training controls, together with inference-time coordinate interventions, show that reaction coordinates change the generative distribution, although the aza-Michael-token intervention was unresolved and individual architectural modules did not improve every family. The finite-catalogue comparison exposed the intended tradeoff: the catalogue often had higher transform-consistent yield, whereas FORGE produced open-ended exact-L1 products and far larger effective component counts. Native recovery of specifically reserved held components remained weak at 0.11 products per 1,000 attempts.

The evidence is strongest for AGILE-type Ugi 3-CR; repeated aza-Michael addition is a second computational family, and repeated reductive amination is a lighter stress test. The study is computational, supports only three qualified assembly programs and operates within a bounded molecular graph space. Exact route closure does not establish experimental synthesis success, and the bounded HeLa diagnostic does not establish biological activity. Neither synthesis tilting nor the nested potency proposal improved its matched terminal endpoint, so both remain negative diagnostics rather than advertised model capabilities. Within these limits, reaction programs provide a practical coordinate system for expanding molecular identity without disconnecting generative design from assembly chemistry.

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A EXTENDED COMPUTATIONAL REPORTING TABLES

The following tables define the complete reporting contract for the common Ugi benchmark. Completed matched controls, the learned inventory selector and three external generators are populated from hash-pinned three-seed results. The method-blind L2/L3 evidence assessment is reported in Table 6.

Table 8: Training and sampling parity for the common Ugi benchmark. Generator calls and attempted products are reported separately from reaction-predictor, verifier, planner, and oracle calls. A method is not described as compute-matched when any required quantity is unavailable. N/R means not recorded by the frozen run; N/P means not applicable.

Method	Params.	Train steps	GPU-hours	Sample steps	Attempts	Reaction calls	Route calls	Oracle calls
RGFN	N/R	5,002 iter.	N/R	N/R	$3 \times 3,072$	0	0	0
DeFoG unconditional	N/R	50,000	N/R	1,000	$3 \times 3,072$	0	0	0
GenMol/SAFE	N/R	50,000	N/R	N/R	$3 \times 3,072$	0	0	0
Learned inventory selector	170,855	2,000	N/R	N/P	$3 \times 3,072$	$3 \times 3,072$	0	0
Finite catalogue oracle	N/P	N/P	N/R	N/P	$3 \times 3,072$	N/R	0	0
Shared-null + post-hoc	N/R	N/R	N/R	N/R	$3 \times 3,072$	N/R	0	0
Input-only program	5,328,301	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
No role loss	5,328,301	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
No core loss	5,328,301	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
No routed adapters	5,026,843	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
No gradient conflict control	5,328,301	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
FACT-matched	5,328,301	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
FACT-generous	29,395,993	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0
FORGE Transformer	5,328,301	1,700	N/R	N/R	$3 \times 3,072$	N/R	0	0

Table 9: Seed-level common Ugi benchmark results. Rows are expanded once a method has a hash-verified three-seed result vector. The main table reports the corresponding mean $\pm$ sample standard deviation; generated molecules are not treated as independent training replicates.

Method	Seed(s)	Valid/1k	Exact L1/1k	Unique L1/1k	Open-ended/1k	Held-comp. L1/1k	Diversity
RGFN	20260825	0.0	0.0	0.0	0.0	0.0	N/E
RGFN	20260826	0.0	0.0	0.0	0.0	0.0	N/E
RGFN	20260827	0.0	0.0	0.0	0.0	0.0	N/E
DeFoG unconditional	20260825	299.2	251.0	251.0	251.0	13.7	0.711
DeFoG unconditional	20260826	359.7	288.4	288.4	288.4	18.2	0.733
DeFoG unconditional	20260827	380.5	302.4	302.4	302.4	14.6	0.736
GenMol/SAFE	20260825	627.3	0.0	0.0	0.0	0.0	0.883
GenMol/SAFE	20260826	633.8	0.0	0.0	0.0	0.0	0.869
GenMol/SAFE	20260827	621.4	0.0	0.0	0.0	0.0	0.866
Learned inventory selector	20260825	1000.0	1000.0	863.0	0.0	0.0	0.717
Learned inventory selector	20260826	1000.0	1000.0	835.6	0.0	0.0	0.717
Learned inventory selector	20260827	1000.0	1000.0	849.9	0.0	0.0	0.715
Finite catalogue oracle	20260825	1000.0	1000.0	841.5	0.0	0.0	0.719
Finite catalogue oracle	20260826	1000.0	1000.0	843.1	0.0	0.0	0.719
Finite catalogue oracle	20260827	1000.0	1000.0	828.1	0.0	0.0	0.719
Shared-null + post-hoc	20260825	183.9	114.3	109.7	109.7	0.3	0.738
Shared-null + post-hoc	20260826	187.5	103.5	98.3	98.3	0.7	0.763
Shared-null + post-hoc	20260827	212.6	125.3	117.5	117.5	0.3	0.761
FACT-matched	20260825	59.2	49.5	48.8	48.8	0.0	0.676
FACT-matched	20260826	81.7	68.7	66.1	66.1	0.0	0.658
FACT-matched	20260827	41.3	37.8	37.4	37.4	0.0	0.668
FACT-generous	20260825	391.0	161.8	161.1	161.1	0.3	0.782
FACT-generous	20260826	747.1	508.8	488.3	488.3	0.0	0.768
FACT-generous	20260827	363.9	219.4	215.5	215.5	0.0	0.771
FORGE Transformer	20260825	801.4	722.7	702.1	702.1	0.0	0.617
FORGE Transformer	20260826	969.1	903.0	843.8	843.8	0.3	0.613
FORGE Transformer	20260827	827.5	734.7	707.7	707.7	0.0	0.634

Table 10: **Reaction decomposition outcomes under the common Ugi verifier.** Coverage is conditioned on molecular validity; replay precision is conditioned on returned decompositions. Abstention and ambiguity retain their prespecified denominators. N/E denotes a conditional metric with no eligible valid product or returned decomposition.

Method	Valid products	Decomp. coverage	Replay precision	Abstention	Ambiguity
RGFN	0.0 $\pm$ 0.0	N/E	N/E	N/E	N/E
DeFoG unconditional	1064.3 $\pm$ 129.9	81.2 $\pm$ 2.4	100.0 $\pm$ 0.0	18.8 $\pm$ 2.4	0.0 $\pm$ 0.0
GenMol/SAFE	1927.7 $\pm$ 19.0	0.0 $\pm$ 0.0	N/E	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Learned inventory selector	3072.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Finite catalogue oracle	3072.0 $\pm$ 0.0	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
Shared-null + post-hoc	598.0 $\pm$ 47.9	58.8 $\pm$ 3.5	100.0 $\pm$ 0.0	41.2 $\pm$ 3.5	0.0 $\pm$ 0.0
FACT-matched	186.7 $\pm$ 62.1	86.3 $\pm$ 4.4	100.0 $\pm$ 0.0	13.7 $\pm$ 4.4	0.0 $\pm$ 0.0
FACT-generous	1538.0 $\pm$ 656.9	56.6 $\pm$ 13.7	100.0 $\pm$ 0.0	43.4 $\pm$ 13.7	0.0 $\pm$ 0.0
FORGE Transformer	2660.3 $\pm$ 277.1	90.7 $\pm$ 2.2	100.0 $\pm$ 0.0	9.3 $\pm$ 2.2	0.0 $\pm$ 0.0

Table 11: **Finite component inventories and source-product coverage.** Inventory size counts unique constitutional components visible to the method at generation time. Product coverage requires that every required role component is present in that inventory. The held-component fold is never used to construct an inventory.

Method	Amine	Aldehyde	Isocyanide	Train coverage	Calibration coverage	Held-comp. coverage
RGFN	185	75	35	100.0	0.0	0.0
DeFoG unconditional	N/P	N/P	N/P	N/P	N/P	N/P
GenMol/SAFE	N/P	N/P	N/P	N/P	N/P	N/P
Finite catalogue oracle	185	75	35	100.0	0.0	0.0
Learned inventory selector	185	75	35	100.0	0.0	0.0
Shared-null + post-hoc	N/P	N/P	N/P	N/P	N/P	N/P
FACT-matched	N/P	N/P	N/P	N/P	N/P	N/P
FACT-generous	N/P	N/P	N/P	N/P	N/P	N/P
FORGE Transformer	N/P	N/P	N/P	N/P	N/P	N/P

B THEORETICAL PROPERTIES

This appendix establishes five properties of FORGE. We show that the sparse state space represents every connected labeled graph within the declared vocabulary, size, and cycle-rank support; characterize the reachability ceiling of a finite-component assembler; prove invariance and population consistency for program-masked denoising; quantify the efficiency of post-hoc rejection under a fixed sampling budget; and establish termination and soundness for the bounded recursive route assessor.

These results concern representation, population objectives, and computational verification. They do not imply that finite-data training assigns positive probability to every representable graph, that registered transformations cover arbitrary chemistry, that bounded route search is complete, or that a returned dossier guarantees experimental synthesis success.

B.1 NOTATION AND SETUP

All graphs in this appendix are finite, connected, undirected, loop-free labeled graphs. Let Σ_V and Σ_E be the finite atom-state and bond-state vocabularies, and define

$$\mathfrak{G}_{N,K} = \{G = (V, E) : |V| \leq N, \beta(G) = |E| - |V| + 1 \leq K, \ell_V(V) \subseteq \Sigma_V, \ell_E(E) \subseteq \Sigma_E\}. \quad (21)$$

Graphs are identified up to label-preserving isomorphism.

A sparse FORGE state with n active nodes is well formed when:

1. $1 \leq n \leq N$, and every active node has a label in Σ_V ;
2. the root has no parent and every node $i > 1$ has one parent $\pi_i < i$;
3. every parent edge has a label in Σ_E ;
4. it contains $k \leq K$ distinct closure edges, each carrying a label in Σ_E , joining two active nodes, and duplicating neither a tree edge nor another closure edge; and
5. closure edges are ordered lexicographically by their endpoint pairs.

Let $\mathfrak{X}_{N,K}$ denote this state space and $\text{Dec} : \mathfrak{X}_{N,K} \rightarrow \mathfrak{G}_{N,K}$ its decoder.

B.2 SPARSE GRAPH REPRESENTATION

Lemma B.1 (Tree residual count). *Let $G = (V, E)$ be a connected finite graph and let T be any spanning tree of G . Then*

$$|E \setminus E(T)| = |E| - |V| + 1 = \beta(G). \quad (22)$$

Proof. A spanning tree on $|V|$ vertices contains exactly $|V| - 1$ edges. Because $E(T) \subseteq E$,

$$|E \setminus E(T)| = |E| - |E(T)| = |E| - (|V| - 1) = |E| - |V| + 1. \quad (23)$$

The final expression is the cyclomatic number of a connected graph. $\square$

Theorem B.2 (Bounded representational completeness). *The decoder*

$$\text{Dec} : \mathfrak{X}_{N,K} \longrightarrow \mathfrak{G}_{N,K} \quad (24)$$

is surjective up to label-preserving graph isomorphism. Equivalently:

1. *for every $G \in \mathfrak{G}_{N,K}$, there exists $X \in \mathfrak{X}_{N,K}$ such that $\text{Dec}(X) \cong G$; and*
2. *every $X \in \mathfrak{X}_{N,K}$ decodes to a graph in $\mathfrak{G}_{N,K}$.*

Proof. Let $G = (V, E) \in \mathfrak{G}_{N,K}$, and choose a spanning tree $T \subseteq G$. Choose a root of T and enumerate its vertices $v_1, \dots, v_n$ in any root-first traversal. Every non-root vertex then appears after its parent.

Assign node i the label of v_i . For each $i > 1$, let $\pi_i < i$ be the index of the parent of v_i in T , and assign the corresponding tree-edge label. By Lemma B.1, the residual edge set $E \setminus E(T)$ contains $\beta(G) \leq K$ edges. Orient each residual edge by increasing endpoint index, attach its original edge label, and sort the resulting triples lexicographically by endpoint pair. The resulting state is an element of $\mathfrak{X}_{N,K}$.

The decoder reconstructs every edge of T from the parent pointers and every edge of $E \setminus E(T)$ from the closure triples. It also preserves all node and edge labels. The map $i \mapsto v_i$ is therefore a label-preserving isomorphism from $\text{Dec}(X)$ to G .

Conversely, let $X \in \mathfrak{X}_{N,K}$ have n active nodes. Since $\pi_i < i$, repeatedly following parent pointers from any node strictly decreases its index and must terminate at the root. The $n - 1$ parent edges therefore form a connected acyclic graph, hence a spanning tree. If the state contains $k \leq K$ closure edges, the decoded graph has

$$|E| = (n - 1) + k \quad (25)$$

distinct edges. Its cyclomatic number is consequently

$$|E| - |V| + 1 = (n - 1 + k) - n + 1 = k \leq K. \quad (26)$$

The remaining well-formedness conditions preserve the declared node and edge vocabularies and exclude loops or duplicate edges. Thus $\text{Dec}(X) \in \mathfrak{G}_{N,K}$. $\square$

B.3 FINITE-COMPONENT REACHABILITY

Theorem B.3 (Finite-component reachability). *Assume that $\mathfrak{T}_P(\mathcal{C})$ is finite and that every forward-outcome set is finite. Then*

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq \sum_{\mathbf{C} \in \mathfrak{T}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|. \quad (27)$$

If $\mathfrak{T}_P(\mathcal{C}) \neq \emptyset$ and

$$M_P(\mathcal{C}) = \max_{\mathbf{C} \in \mathfrak{T}_P(\mathcal{C})} |\mathcal{F}_P(\mathbf{C})|, \quad (28)$$

then

$$|\mathcal{G}_{\text{cat}}(P, \mathcal{C})| \leq M_P(\mathcal{C}) |\mathfrak{T}_P(\mathcal{C})|. \quad (29)$$

Moreover,

$$G \in \mathcal{G}_{\text{cat}}(P, \mathcal{C}) \iff \mathcal{E}_P^*(G) \cap \mathfrak{T}_P(\mathcal{C}) \neq \emptyset. \quad (30)$$

Every catalogue attempt uses only catalogue components, so its attainable component-novelty fraction is exactly zero.

Proof. The reachable set is a finite union of forward-outcome sets. The cardinality of a union is at most the sum of the cardinalities of its members, which proves the first inequality. Bounding each summand by $M_P(\mathcal{C})$ proves the second.

By definition, G belongs to the union precisely when at least one $\mathbf{C} \in \mathfrak{T}_P(\mathcal{C})$ satisfies $G \in \mathcal{F}_P(\mathbf{C})$. This is equivalent to membership in $\mathcal{E}_P^*(G) \cap \mathfrak{T}_P(\mathcal{C})$. Finally, every component in an input from $\mathfrak{T}_P(\mathcal{C})$ belongs to its role-specific catalogue by construction, so component novelty is impossible. $\square$

B.4 PROGRAM-MASKED DISCRETE FLOW

For a program P , let $\mathcal{I}(P)$ index the categorical coordinates of its sampling layout. Define

$$\mathcal{S}(P) = \left\{ (q, j) \in \mathcal{I}(P) : m_j^{(q)}(P) = 0 \right\} \quad (31)$$

and let $x_{P,j}^{(q)}$ be the prescribed value of each fixed coordinate. The program-compatible face of the raw categorical state space is

$$\Omega_P = \left\{ x : x_j^{(q)} = x_{P,j}^{(q)} \text{ for every } (q, j) \in \mathcal{S}(P) \right\}. \quad (32)$$

Define the projection Π_P coordinatewise by

$$[\Pi_P(x)]_j^{(q)} = \begin{cases} x_{P,j}^{(q)}, & (q, j) \in \mathcal{S}(P), \\ x_j^{(q)}, & (q, j) \notin \mathcal{S}(P). \end{cases} \quad (33)$$

Training corruption has the form $X_t = \Pi_P(\tilde{X}_t)$. Numerical sampling begins from $X_0 = \Pi_P(\tilde{X}_0)$ and applies

$$X_{k+1} = \Pi_P(\Phi_k^\theta(X_k, \xi_k)), \quad k = 0, \dots, L-1, \quad (34)$$

where Φ_k^θ is the unconstrained stochastic update and ξ_k contains its sampling randomness.

Proposition B.4 (Program-fixed invariance). *For every program P ,*

$$\Pr[X_t \in \Omega_P] = 1 \quad (35)$$

at every corruption time t , and

$$\Pr[X_k \in \Omega_P \text{ for every } k = 0, \dots, L] = 1 \quad (36)$$

during numerical sampling.

Proof. By definition, $\Pi_P(x) \in \Omega_P$ for every raw state x , and $\Pi_P(x) = x$ for every $x \in \Omega_P$. Since corruption returns $\Pi_P(\tilde{X}_t)$, every corrupted state lies in Ω_P .

The sampling claim follows by induction. The projected initial state belongs to Ω_P . If $X_k \in \Omega_P$, the unconstrained update may change any coordinate, but applying Π_P restores every coordinate in $\mathcal{S}(P)$. Hence $X_{k+1} \in \Omega_P$. The conclusion holds for all sampling steps. $\square$

Define the joint population law

$$\nu(dG, dP, dt, dX_t) = \mu_{\text{bal}}(dG, dP) \mathcal{U}(dt) p_t(dX_t | X_G, P). \quad (37)$$

For a variable coordinate, let

$$\rho_{qj}(a | X_t, t, P) = \Pr[X_{G,j}^{(q)} = a | X_t, t, P]. \quad (38)$$

Theorem B.5 (Population optimality of masked denoising). *Assume an unrestricted conditional model class and $\lambda_q > 0$. For every coordinate satisfying $m_j^{(q)}(P) = 1$, every population minimizer of $\mathcal{L}_{\text{FORGE}}$ satisfies*

$$p_\theta^*(X_{G,j}^{(q)} = a | X_t, t, P) = \rho_{qj}(a | X_t, t, P) \quad (39)$$

for every categorical state a , ν -almost surely. Coordinates with $m_j^{(q)}(P) = 0$ contribute no loss and are instead governed by Proposition B.4.

Proof. Fix a family q , a variable coordinate j , and a context $C = (X_t, t, P)$ for which the regular conditional distribution is defined. Let $p(a) = p_\theta(X_{G,j}^{(q)} = a | C)$ and $\rho(a) = \rho_{qj}(a | C)$. The conditional expected loss is

$$-\sum_a \rho(a) \log p(a) = -\sum_a \rho(a) \log \rho(a) + \sum_a \rho(a) \log \frac{\rho(a)}{p(a)} \quad (40)$$

$$= H(\rho) + D_{\text{KL}}(\rho || p). \quad (41)$$

The entropy term is independent of p . By non-negativity of Kullback–Leibler divergence, the expression is minimized exactly when $p = \rho$.

For fixed P and q , the factor $\lambda_q / \sum_j m_j^{(q)}(P)$ is positive and independent of the predicted categorical distribution. It therefore does not change the pointwise minimizer. Integrating over contexts and repeating the argument for every variable coordinate proves the result. Coordinates with zero masks do not appear in the objective. $\square$

B.5 MATCHED-BUDGET POST-HOC FILTERING

Fix a reaction program P , and define

$$A_P(G) = \mathbb{1}[G \in \mathcal{G}_{L1}(P)]. \quad (42)$$

For a proposal distribution Q on the bounded graph space, let

$$Z_Q(P) = \Pr_{G \sim Q}[A_P(G) = 1] = \mathbb{E}_{G \sim Q}[A_P(G)] \quad (43)$$

denote the exact-L1 acceptance probability.

Proposition B.6 (Exact law of matched-budget post-hoc filtering). *Suppose $Z_Q(P) > 0$. Rejection filtering produces the conditional distribution*

$$Q_{\text{post}}(g \mid P) = \Pr_{G \sim Q}[G = g \mid A_P(G) = 1] = \frac{Q(g)A_P(g)}{Z_Q(P)}. \quad (44)$$

For B independent attempts $G_1, \dots, G_B \sim Q$, define

$$N_B = \sum_{b=1}^B A_P(G_b). \quad (45)$$

Then

$$N_B \sim \text{Binomial}(B, Z_Q(P)), \quad (46)$$

and consequently

$$\mathbb{E}[N_B] = BZ_Q(P), \quad (47)$$

$$\text{Var}(N_B) = BZ_Q(P)(1 - Z_Q(P)), \quad (48)$$

$$\Pr[N_B = 0] = (1 - Z_Q(P))^B. \quad (49)$$

If T_K is the number of independent attempts required to obtain K accepted products, then

$$\mathbb{E}[T_K] = \frac{K}{Z_Q(P)}. \quad (50)$$

Finally, for a program-conditioned proposal Q_P and an unconditioned proposal Q_0 ,

$$\mathbb{E}[N_B^P] - \mathbb{E}[N_B^0] = B(Z_{Q_P}(P) - Z_{Q_0}(P)). \quad (51)$$

Proof. For any graph g , Bayes' rule gives

$$\Pr[G = g \mid A_P(G) = 1] = \frac{\Pr[A_P(G) = 1 \mid G = g] \Pr[G = g]}{\Pr[A_P(G) = 1]} \quad (52)$$

$$= \frac{A_P(g)Q(g)}{Z_Q(P)}. \quad (53)$$

Since the proposals are independent, the indicators $A_P(G_1), \dots, A_P(G_B)$ are independent Bernoulli random variables with common success probability $Z_Q(P)$. Their sum is therefore binomial, which gives its expectation, variance, and zero-acceptance probability.

The waiting time T_K has the negative-binomial distribution with success probability $Z_Q(P)$, whose expectation is $K/Z_Q(P)$. Applying the binomial expectation separately to Q_P and Q_0 and subtracting proves the final identity. $\square$

The proposition compares sampling efficiency only. It does not imply that conditioning improves $Z_Q(P)$; the matched production experiment tests that empirical claim.

B.6 RECURSIVE ROUTE CONSTRUCTION

Lemma B.7 (Termination of bounded route assessment). *Fix an assessment contract $\mathcal{B}$ with finite maximum route depth $D_{\mathcal{B}}$. Assume that each evidence query terminates and returns either a non-expansion outcome or an expansion with finitely many reactants. Then $\mathcal{A}_{\mathcal{B}}(G, P)$ terminates with either a finite route tree or a typed failure state.*

Proof. We use induction on the remaining depth $d = D_{\mathcal{B}} - k$ at a node of depth k . When $d = 0$, the assessor performs the terminating evidence query and either accepts a terminal node, returns its typed failure, or rejects an expansion with a depth-exhaustion state. Suppose the claim holds whenever fewer than $d > 0$ levels remain. At the current node, the evidence query either returns a non-expansion outcome immediately or a finite reactant tuple. In the latter case, each child is assessed at depth $k + 1$ and therefore has $d - 1$ levels remaining. Every child assessment terminates with a finite result by the induction hypothesis, and a finite tuple of finite child trees forms a finite parent tree. Finite expansion, proposal, verifier, and time budgets can only stop this recursion earlier. Rejecting an ancestor identity additionally prevents a circular molecular dependency from being accepted, but finite depth alone establishes termination. $\square$

Theorem B.8 (Soundness of returned synthesis dossiers). *Fix $\mathcal{B}$ as in Lemma B.7. If $\mathcal{A}_{\mathcal{B}}(G, P) = T$, then T is finite and $\text{Complete}(G, T, P) = 1$. In particular, G passes exact L1 verification under P , every L2 reaction in T reconstructs its target exactly under its registered forward transform, and every leaf has a current accepted record identifying the same constitutional structure as supplier-listed and available. A candidate tree with $\text{Complete}(G, T, P) = 0$ cannot be returned as a synthesis dossier. The outcome $\mathcal{A}_{\mathcal{B}}(G, P) = \perp$ states only that no complete dossier was returned under $\mathcal{B}$. It does not prove that no route exists under a larger search budget, a different admitted reaction set, or a later supplier snapshot.*

Proof. Finiteness follows from Lemma B.7. It remains to establish the three factors in the completeness predicate. The assessor begins recursive route construction only after exact L1 decomposition and forward replay have succeeded, so a returned dossier satisfies $\text{L1}(G, P) = 1$.

We next use structural induction on the finite route tree. At a leaf, the assessor returns a complete node only when the evidence policy admits an exact terminal-material record whose availability state is current and closed. Therefore $\text{Available}(\ell) = 1$ for every leaf ℓ of a returned tree. At an internal node, the assessor admits an expansion only when the registered reaction has exact-source, exact-substrate evidence and its deterministic forward transform uniquely reconstructs the target product. Thus $\text{Verify}(r) = 1$ for the associated reaction node r . The internal node is complete only when every child is complete; the induction hypothesis therefore applies to all descendant reactions and leaves.

All factors in $\text{Complete}(G, T, P)$ equal one, so their product equals one. Conversely, if any factor for a candidate tree equals zero, the fail-closed return rule does not admit that tree as a dossier and instead records the corresponding typed failure. Because the search and evidence contract is bounded, this rejection establishes no completeness claim about routes outside $\mathcal{B}$. $\square$

C GENERATED-STRUCTURE ATLAS

Table 12 reports 12 deterministic display examples from the same frozen seed-0 Ugi population used in Figure 3. The display begins with the lowest constitutional SHA-256 rank and then applies ECFP4 MaxMin selection among the 23 eligible, constitutionally unique products. This promotes visual coverage without using activity, route closure, procurement evidence or human preference. Each row includes the canonical constitutional SMILES so that the displayed identity is explicit; the exact strings are also stored in the hash-pinned atlas receipt.

Table 12: **FORGE generated-structure atlas**. The 12 rows form one multipage table, indexed by display order. Each constitutional graph is followed by its canonical SMILES. The decorated three-dimensional column shows one deterministic post-hoc RDKit ETKDgV3 structure optimized with up to 400 MMFF94s iterations. Purple denotes the amine-derived head, teal the aldehyde-derived body and tail, orange the isocyanide-derived tail and gray the assembly-introduced oxygen; these assignments are recovered through exact atom-mapped forward replay. FORGE does not generate three-dimensional coordinates, and these views are not predicted bioactive conformations. Exact L1 replay and descriptor-manifold membership are structural diagnostics, not evidence of activity, route closure or synthesis success. These are deterministic display examples, not experimental candidates.

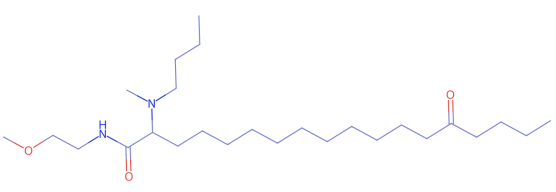
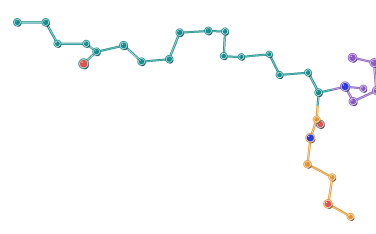
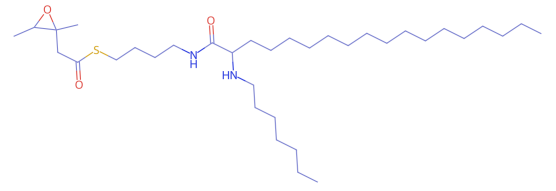
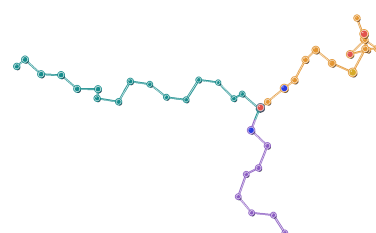
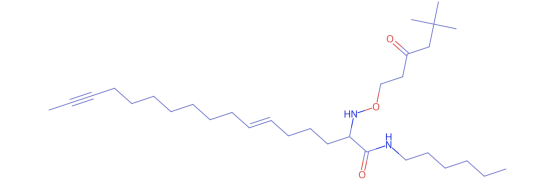
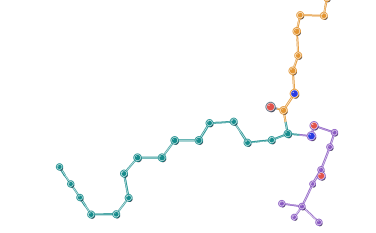
CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
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2  <chem>CCCCCCCCCCCCCCCC(NCCCCC)C(=O)NCCCSC(=O)CC1(C)OC1C</chem>	
3  <chem>CC#CCCCCCCCC=CCCCC(NOCCC(=O)CC(C)(C)C(=O)NCCCCC</chem>	

Table 12, continued

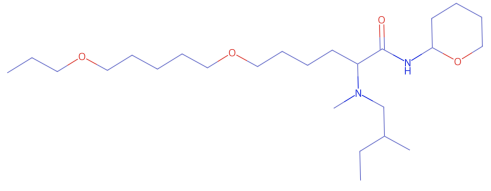
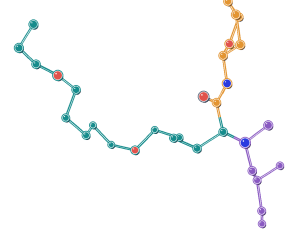
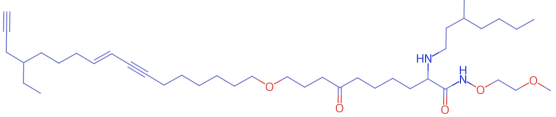
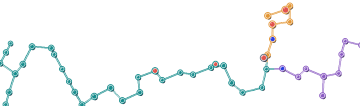
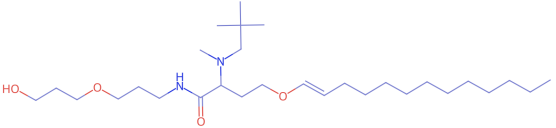
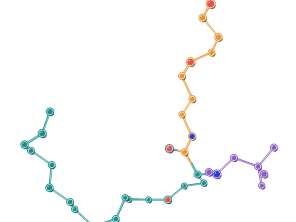
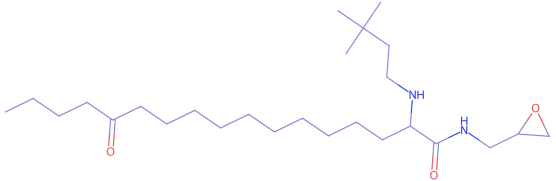
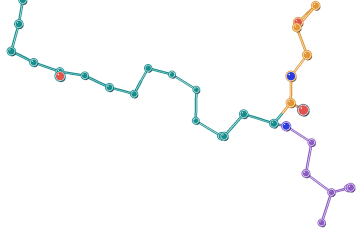
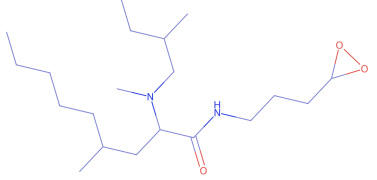
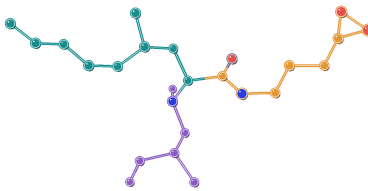
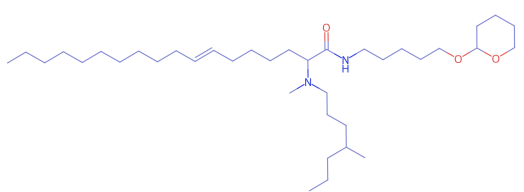
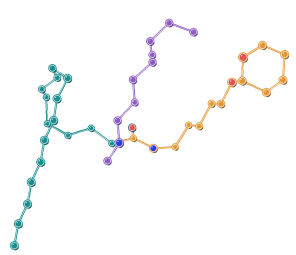
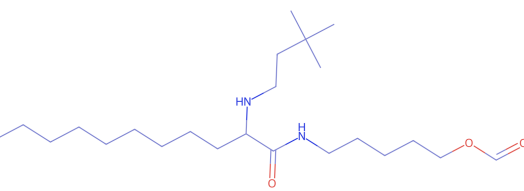
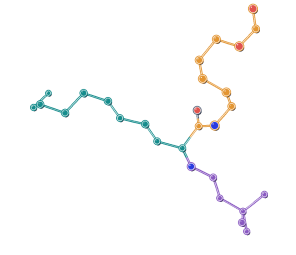
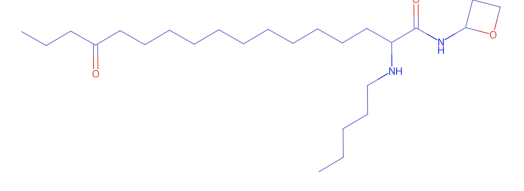
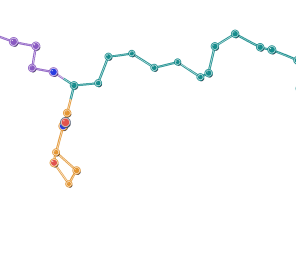
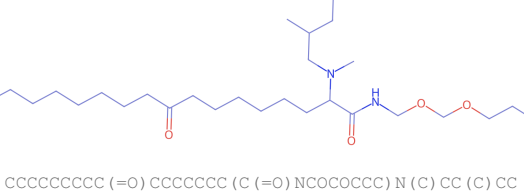
CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
4  <chem>CCCCCCCCCCCCC(C(=O)NC1CCCCO1)N(C)CC(C)CC</chem>	
5  <chem>C#CCC(CC)CCCC=CC#CCCCCCCCCCCCC(=O)CCCC(NCCC(C)CCCC)C(=O)NOCCOC</chem>	
6  <chem>CCCCCCCCCCCCC=CCCC(C(=O)NCCC(=O)CO)N(C)CC(C)(C)C</chem>	
7  <chem>CCCCC(=O)CCCCCCCCCCCC(NCCC(C)(C)C)C(=O)NCC1CO1</chem>	
8  <chem>CCCCC(C)CC(C(=O)NCCCC1OO1)N(C)CC(C)CC</chem>	

Table 12, continued

	CONSTITUTIONAL GRAPH AND CANONICAL SMILES	DECORATED 3D VIEW
9	 <chem>CCCCCCCCC=CCCCC(C(=O)NCCC(COC1CCCCO1)N(C)CCCC(C)CC</chem>	
10	 <chem>CCCCCCCCC(NCCC(C)(C)C)C(=O)NCCCCOC=O</chem>	
11	 <chem>CCCCCNC(CCCCCCCCCC(=O)CCC)C(=O)NC1CCO1</chem>	
12	 <chem>CCCCCCCCC(=O)CCCCC(C(=O)NCOC(CCC)N(C)CC(C)CC</chem>	